SAUDI WATER AUTHORITY — INTERNAL PLATFORM DOCUMENTATION

Tadawul Platform

Technical & UX Reference — Task Management System

Version: 1.0 Date: May 2026 Audience: Development Team Status: Draft Sample

1. System Overview

The **Task Management System** (اجتماع تداول) is the core workflow module of the Tadawul platform. It enables cross-departmental task creation, assignment, tracking, and approval — all within a role-based access model that reflects the organizational hierarchy of the Saudi Water Authority.

1.1 User Roles

Admin / مشرف

Full visibility across all departments. Can create, edit, delete and approve any task.

Deputy / وكيل

Cross-department read/write. Creates tasks, approves change requests, exports reports.

Manager / مدير

Department-scoped. Manages own department's tasks, approves employee submissions.

Employee / موظف

Personal task view. Updates status on assigned tasks, adds comments, submits change requests.

2. UX Walkthrough

2.1 Landing / Hub Page

After login, all users land on a unified hub page that surfaces the systems they are authorised to access. The hub adapts to the user's role — an employee sees only the systems relevant to them, while an admin sees all five. The topbar shows the user's name, role, last login time, and server status indicator.

Figure 1 — Hub landing page (Admin view, showing all 5 systems)

2.2 Deputy / Admin View

The deputy/admin dashboard provides a **full cross-department view**. Key elements:

- **Department tabs** — filter tasks by department with live counts per tab.
- **KPI cards** — at-a-glance counts for pending approval, high priority, in-progress, and completed tasks.
- **Approval queue** — dedicated panel showing tasks awaiting sign-off, separated from the main list.
- **Stopwatch widget** — floating meeting timer (draggable) for use during weekly review sessions.
- **Export** — one-click Excel export of the current filtered view.
- **Kanban / Table toggle** — switch between tabular and board-style task views.

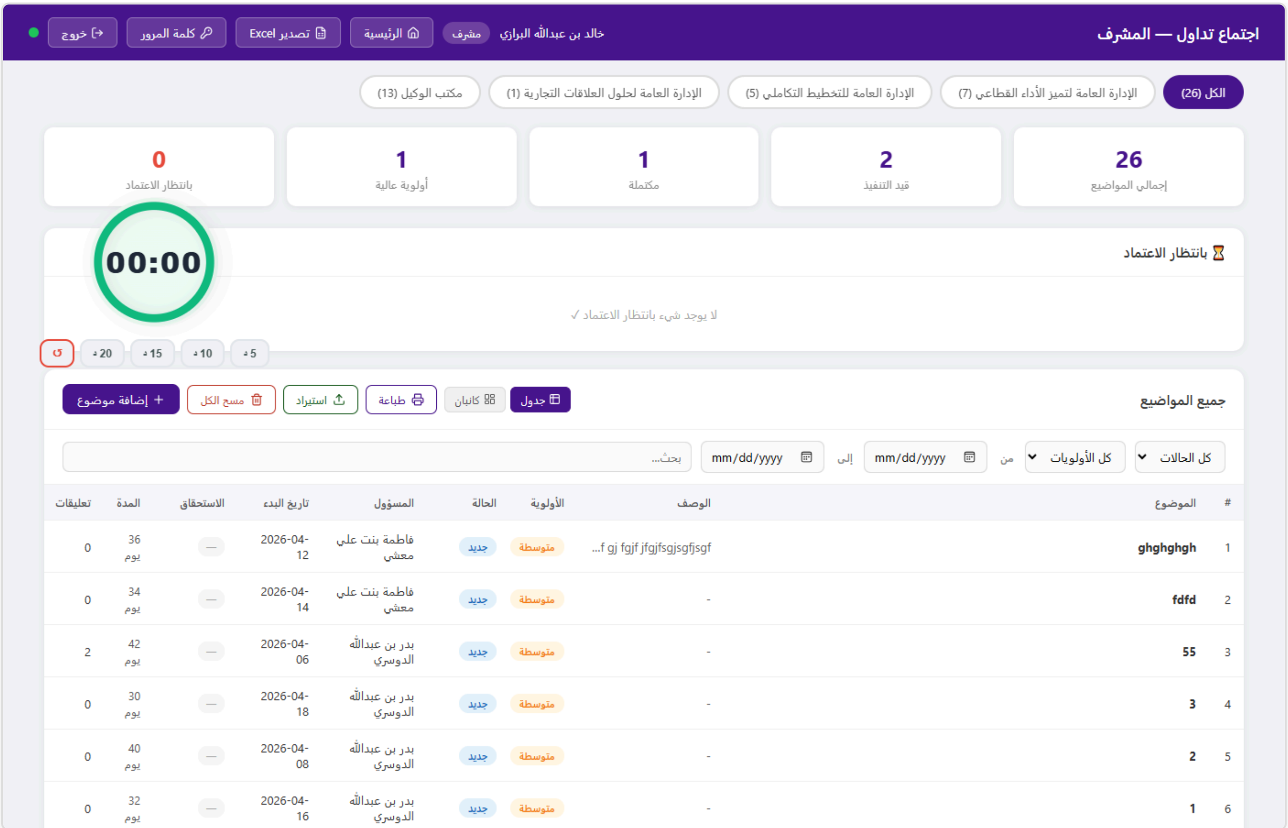

Figure 2 — Deputy/Admin task dashboard showing cross-department tasks, KPI cards, and the approval queue

2.3 Task Creation

Tasks are created via a modal dialog accessible to managers, deputies, and admins. Fields include title, description, target department, assignee, priority, start date, and due date. On save, the task is immediately visible to the assigned employee and the department manager, with a real-time push notification via Server-Sent Events (SSE).

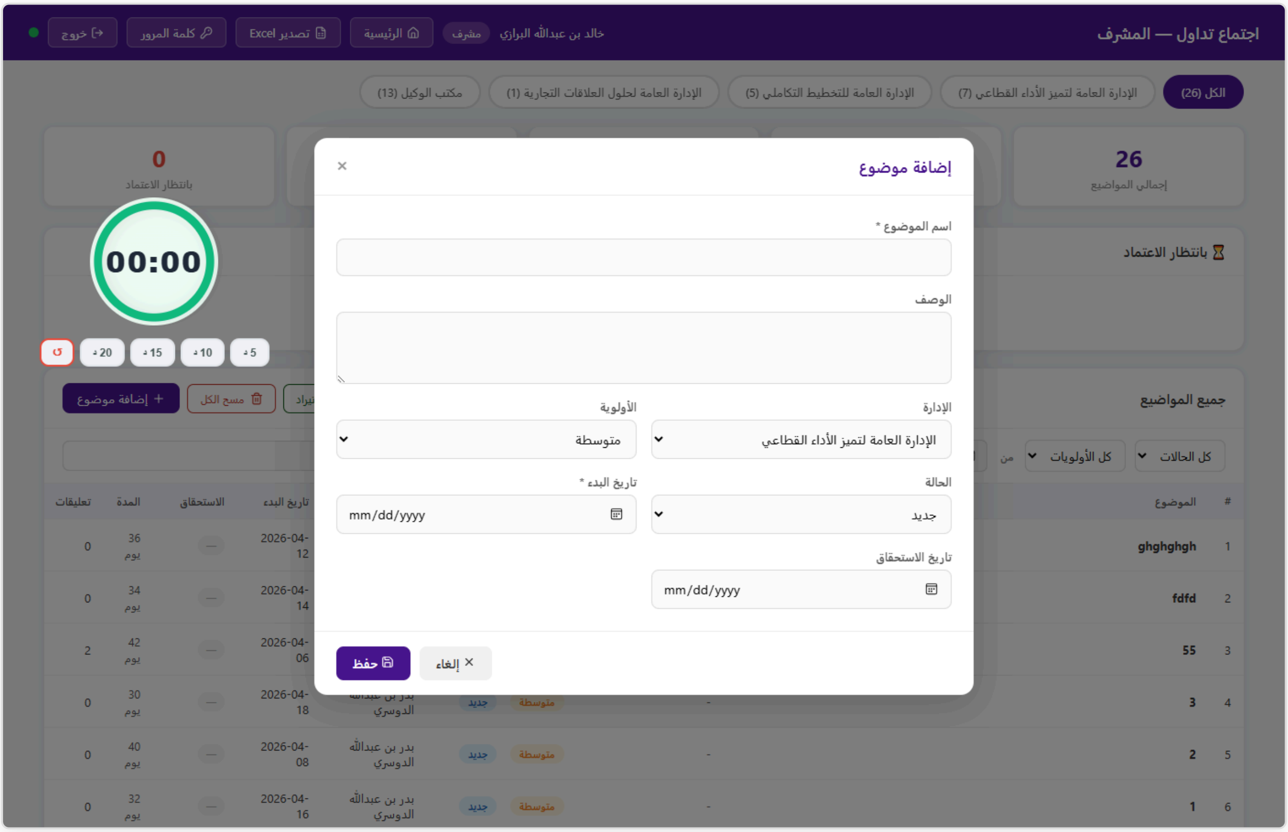

Figure 3 — Task creation modal (Deputy view)

2.4 Manager View

Managers see tasks scoped to their department only. The interface includes an approval queue for employee-submitted change requests (status updates, due date adjustments) and a full task list with filters for status, priority, assignee, and date range.

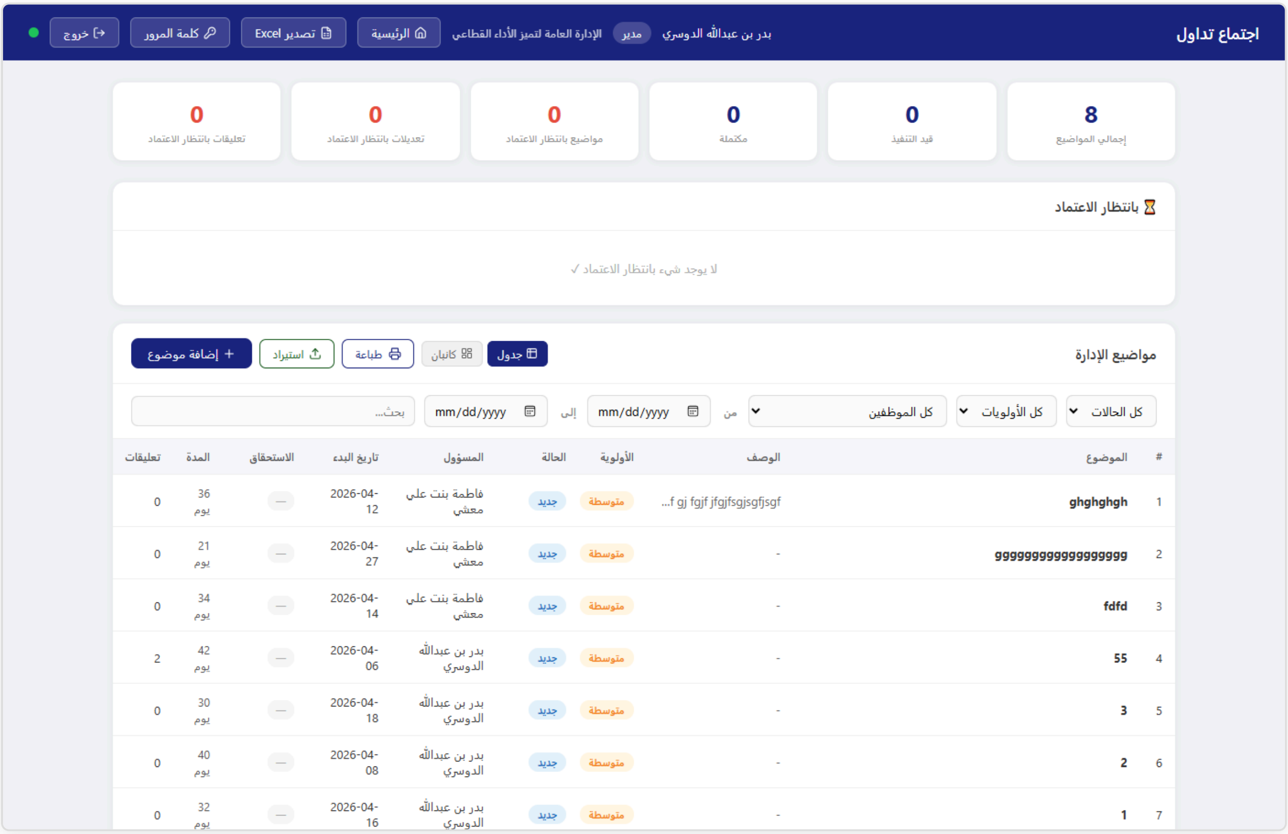

Figure 4 — Manager task dashboard (department-scoped)

2.5 Employee View

Employees see only tasks assigned to them. They can update task status (with manager approval required for certain transitions), add comments/guidance entries, and submit change requests for due dates. The interface intentionally omits cross-department visibility and administrative controls.

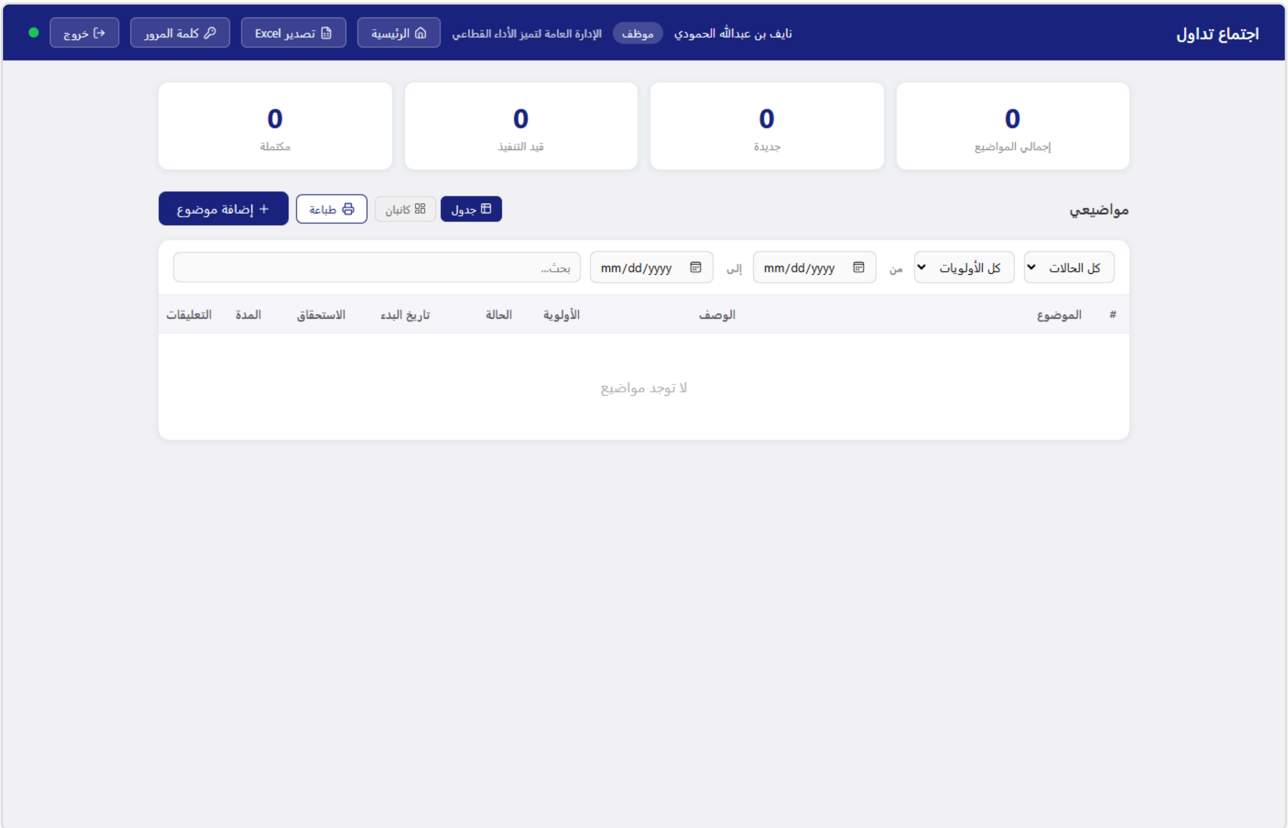

Figure 5 — Employee task view (personal tasks only)

3. Key Business Rules

3.1 Task Status Workflow

جديد New	→	قيد التنفيذ In Progress	→	مكتملة Completed

Status transitions by employees generate change requests that appear in the manager's approval queue. Deputies and admins can transition any task directly without approval.

3.2 Change Request Approval

- Employees submit requests to change status or due date — these do *not* apply immediately.
- The request appears in the manager's and deputy's approval panel.
- On approval, the change is applied and the employee is notified via real-time SSE push.
- On rejection, the task remains unchanged and the request is marked declined.

3.3 Real-Time Synchronisation

> All connected clients receive live updates when any task is created, modified, or approved. This is implemented via **Server-Sent Events (SSE)** — an open HTTP stream from the server that pushes a broadcast signal whenever the data changes. Clients respond by silently re-fetching their data in the background. No manual refresh is needed.

4. Technical Architecture

4.1 Current Stack

Layer	Technology	Purpose
Runtime	Node.js 20 + Express 5	HTTP server, REST API, SSE endpoint
Database	SQLite (better-sqlite3)	All persistent data — tasks, users, departments, history
Authentication	JWT (jsonwebtoken, bcryptjs)	Stateless session tokens, 8-hour expiry, role embedded in token
Real-time	Server-Sent Events (SSE)	One-way server-to-client broadcast on data changes
Frontend	Vanilla HTML/CSS/JS	Static files served by Express; no framework, no build step
Icons	Lucide (CDN)	Consistent icon set across all pages
Export	SheetJS (XLSX, CDN)	Client-side Excel generation from filtered task data

4.2 Data Model — Tasks

Field	Type	Description
id	TEXT (UUID)	Primary key
name	TEXT	Task title
description	TEXT	Optional detail
priority	TEXT	عالية / متوسطة / منخفضة
status	TEXT	جديد / قيد التنفيذ / مكتملة
dept_id	INTEGER	FK → departments
assigned_to	TEXT	FK → employees
created_by / role	TEXT	Creator identity + role at creation time
start_date / due_date	TEXT (ISO)	Date range
approved	INTEGER	1 = live, 0 = pending deputy approval
created_at	TEXT (ISO)	Server timestamp

4.3 API Endpoints — Tasks

Method	Endpoint	Access	Description
GET	/api/tasks/all	Deputy, Admin	All tasks, all departments

GET	/api/tasks/dept/:id	Manager	Tasks for own department
GET	/api/tasks/my	Employee	Tasks assigned to calling user
POST	/api/tasks	Manager+	Create task
PATCH	/api/tasks/:id	Manager+	Update task fields directly
POST	/api/tasks/:id/changes	Employee	Submit change request
PATCH	/api/tasks/changes/:id	Manager+	Approve / reject change request
DELETE	/api/tasks/:id	Deputy, Admin	Delete task

4.4 Authentication Flow

Login POST /api/auth/login	→	**JWT issued** Role + ID embedded	→	**Stored** sessionStorage (tab-scoped)	→	**API calls** Bearer header on every request	→	**Middleware** Validates + injects req.user

4.5 Real-Time Architecture

A single `/api/events` SSE endpoint maintains an open connection to every active browser tab. On any write operation (task create/update/delete/approve), the server calls `broadcast()` which sends a lightweight signal to all connected clients. Each client re-fetches only its own data slice. A server-side heartbeat (every 5 seconds) keeps connections alive and allows clients to detect server unavailability via a 12-second silence threshold.

> **Server status indicator:** A coloured dot in every page's topbar reflects live connectivity — green (server up), red (server unreachable). State is driven by a 3-second HTTP ping and the SSE heartbeat, with no user action required to reconnect after a server restart.